Random Graph

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Random Graph

- We introduce a version of **random graph model** (Erdős and Rényi (1959), Gilbert (1959)).
- This model is useful because...
  - it is a simple model of graph that is characterized by a small number of parameters (just 2).
  - it can be used as a good benchmark, as it represents a completely random network.
  - we can establish various properties asymptotically (when $n \to \infty$).
  - it sheds light on certain phenomena (small-worldness).
- On the other hand, there are many features of social networks that it cannot replicate.

We consider the following random graph model $G(n, p)$ with parameter n and p:

- n nodes.
- Each pair of nodes is linked with probability p independently.

Another possible formulation (still with 2 parameters) would be to fix m and pick m links from $\frac{n(n-1)}{2}$ pairs of nodes uniformly randomly. This graph is similar to (but different from) $G(n, p)$ when $m = \frac{n(n-1)}{2}p$.

Link and Degree

Since network is random, various metrics are random variables. Let's check their properties. We start with degree.

- Let L_i be the number of links for node i.
- i's expected degree is $\lambda = E[L_i] = (n-1)p$ for every i. Average degree is $\frac{\sum_i L_i}{n}$, so the expected average degree is $E[\frac{\sum_i L_i}{n}] = \frac{n(n-1)p}{n} = (n-1)p$ as well.
- $\#$ (all links) is $\frac{\sum_i L_i}{2}$. So, the expected number of links is $E[\frac{\sum_i L_i}{2}] = \frac{n(n-1)p}{2}$.

Degree Distribution

- Let n be arbitrarily large while keeping the expected degree λ fixed (hence $p(n) = \frac{\lambda}{n-1}$). Then the degree distribution at each node converges to **Poisson distribution**: $P^D(d) = \frac{\lambda^d e^{-\lambda}}{d!}$.
  - Note that $\frac{P^D(d+1)}{P^D(d)} = \frac{\lambda}{d+1}$, which decrease at the rate of $\frac{1}{d+1}$. So, no fat tail.

Crustering

- When (i, j) and (j, k) are linked, the probability that i and k are linked is just p by independence.
  - By definition, independence implies there is no triadic closure, which is an important feature of social networks.
  - If we fix λ and make n large (as we did for degree distribution), then $p(n) \to 0$ as $n \to \infty$. So, the clustering coefficient is 0 with probability 1 as $n \to \infty$. To achieve high clustering in large network, we need to set λ unrealistically high.

Average Distance and Diameter

- What is the average distance between any two nodes?
- Assume $\lambda > 1$ and imagine the following process (**Branching process**): Pick any initial node. Generate Z links from this node to Z nodes, where Z is a random variable with $E[Z] = \lambda$. From each node that is generated in the first step, generate new Z links to new Z nodes in the second step and so on (the number of links are i.i.d. across steps). This generates an expanding tree graph.
- For such a process, the expected number of nodes reached in step k is λ^k. Consider k^* such that $\lambda^{k^*} = n$: the expected number of reached nodes is equal to the size of the network.

Average Distance and Diameter

- This $k^*(n) = \frac{\log n}{\log \lambda}$ turns out to be a good approximation of the average distance when n is large (in the sense that $\frac{\text{average distance}(n)}{k^*(n)}$ converges to 1 with probability 1). So, the average distance increases at the rate of $\log n$ for a given λ.
- $\frac{\log n}{\log \lambda}$ is also a good approximation of the diameter of the network.
- What if $\lambda < 1$? In this case, the most of nodes are not reachable from one node for large n as we will see. Intuitively, this is because the expected number of nodes that can be reached from one node is at most $\lambda + \lambda^2 + \lambda^3 + ... = \frac{\lambda}{1-\lambda} < 0$ independent of n.

Asymptotic Properties of $G(n,p)$

- We can establish many nice properties for this random graph when $n \to \infty$. Moreover, we can often find where a **phase transition** occurs and certain properties start to emerge.
- Here are two such properties.
  - If $p(n) < (1-\epsilon)\frac{\log n}{n}$, then the probability of the network being connected goes to 0 as $n \to \infty$. If $p(n) > (1+\epsilon)\frac{\log n}{n}$, then the probability of the network being connected goes to 1 as $n \to \infty$.
  - If $p(n) < (1-\epsilon)\frac{1}{n}$, then the fraction of the largest component converges to 0 as $n \to \infty$. If $p(n) > (1+\epsilon)\frac{1}{n}$, then there would be the unique **giant component** in the network as $n \to \infty$.

Giant Component

- Giant component is a component whose size is proportional to n.
- There cannot be two giant components (with probability $\to 0$) when n is large, because the probability of no link between such two big components must go to 0 as n increases.
- The previous result says that, as $n \to \infty$, there is no giant component when $p < \frac{1}{n}$, the whole network is the giant component when $p > \frac{\log n}{n}$, and the unique giant component occupies a fraction of the network for p between two magic numbers $\frac{1}{n}$ and $\frac{\log n}{n}$.

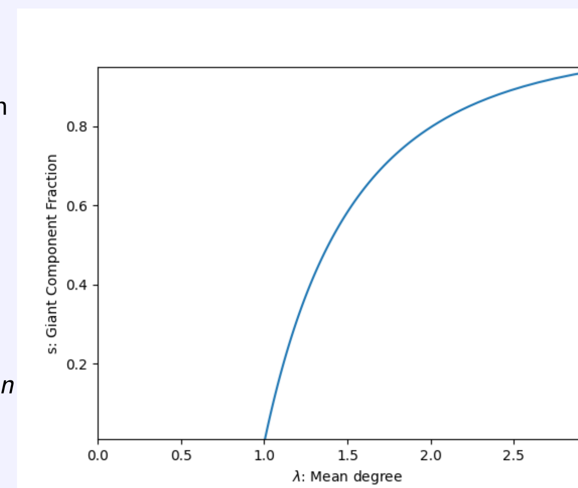
Size of Giant Component

We can do some back of envelope calculation to derive the size of the giant component when $\frac{1}{n} < p < \frac{\log n}{n}$ (as $n \to \infty$).

- Consider a graph with $n-1$ nodes where the fraction of giant component is s.
- Add one node i. For i to be not connected to the giant component through node j, i is either not connected to j or connected to j but j is not connected to the giant component.
- So, the probability of i not being connected to the giant component is $(1-p+p(1-s))^{n-1}$. At the same time, this probability must be $1-s$. Hence $1-s=(1-p+p(1-s))^{n-1}$
- From this, we can get some formula about s: $s = 1-e^{-\lambda s}$, which is valid when n is large relative to λ.

Size of Giant Component

- This equation always has 0 as a solution, which is the only solution fup to $\lambda = 1$, meaning no giant component.
- For $\lambda > 1$, there is a positive solution s, which increases with λ and reach 1 as λ gets closer to $\ln n$ (but it can get close to 1 quickly for a small λ as shown).



Small-World Model

- Random graph model does not replicate certain features of real social networks.
  1. It cannot generate high individual clustering for a reasonable level of p, λ.
  2. The degree distribution follows a Poisson distribution with parameter λ ($=$ expected degree) when n is large. In many social networks, the degree distribution is more fat-tailed than Poisson.
- Watts and Strogatz (1998) proposed a model that keeps high individual clustering while keeping the reasonable average distance level (so, this addresses the first problem).
- The main idea is to combine a highly clustered network with weak ties.

Here is one simplified version of the algorithm.

- Pick a target level of (integer) degree λ. This could be something close to the observed average degree. Or it could be close to λ that satisfies $\lambda^6 = n$ if you want "six degrees of separation".
- Consider a ring graph with n nodes. For each node, add $\lambda - 2$ edges to the closest $\lambda - 2$ nodes except for two adjacent nodes that are already linked (assume λ is an even number for simplification).
- For each node, rewire each of λ edges with probability β independently to any other node uniformly randomly, except that self-link and link-duplication is avoided.

How does this process look like? Try to visualize it...

- When $\beta = 0$ (no rewiring), the graph is λ-regular graph with very high individual clustering.
- When $\beta = 1$, this is a version of random graph where the degree of each node is constant λ, with low individual clustering.
- We can find a right $\beta \in (0, 1)$ to achieve an appropriate level of high individual clustering and short average distance, while keeping the degree to λ.

Summary

- Random graph and its basic properties.
- Two threshold for p as $n \to \infty$.
  - $\frac{\log n}{n}$: connectedness
  - $\frac{1}{n}$: giant component
- Small-World Model: high clustering and short average disance for a reasonable range of parameters.

Suggested reading. EK Ch.20.1, Ch.20.2